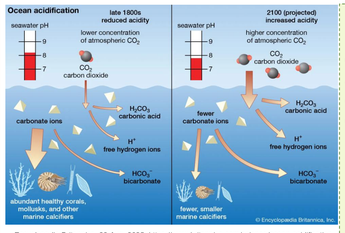
UPISC UNIVERSITY OF PITTSBURGH INFRASTRUCTURE SENSING COLLABORATION WORKSHOP

NATIONAL ENERGY TECHNOLOGY LABORATORY

Distributed fiber optic pH sensing for marine environment monitoring

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Introduction



Encyclopedia Britannica, 28 Aug. 2025, https://www.britannica.com/science/ocean-acidification

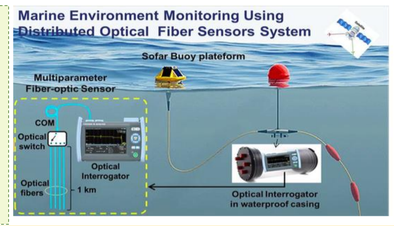
pH in the ocean is a very important parameter to measure as it affects CO_2-carbonate equilibrium system of seawater and ocean acidification. The increased acidity reduces the availability of carbonate ions which are essential for marine calcifiers thereby significantly affecting the marine ecosystems.

1% uncertainty in carbonate ions ⟶ Uncertainty of ~0.003pH units

10% uncertainty in carbonate ions ⟶ Uncertainty of ~0.02pH units

Flexible and compact optical fiber sensors with resistance to electromagnetic interference have great potential for real time in-situ monitoring of chemical and physical parameters in marine environment through distributed interrogation.

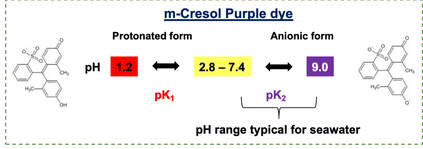
- Distributed fiber optic chemical sensing demands functionalized analyte specific coatings that convert chemical information to detectable optical or mechanical properties



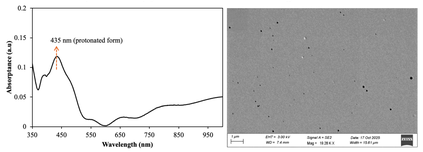
pH sensing layer & characterization

Sensing layer	Indicator dye immobilized in organically modified silica
Mechanism	Change in absorption due to protonation and deprotonation of indicator dye
Advantages	• Low, tunable refractive index depending on porosity • Addition of non hydrolysable organic functional group increase flexibility and hydrophobicity as well as reduce dye leaching • Compatible with marine environment • Solution processible: feasible with reel-to-reel coating for distributed sensing
Compatible interrogator	Optical time domain reflectometry (OTDR): custom wavelength required

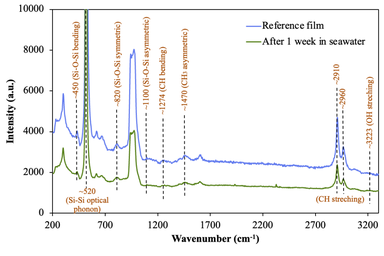
m-Cresol Purple dye



UV-Vis-NIR spectroscopy and SEM surface image of coated thin films



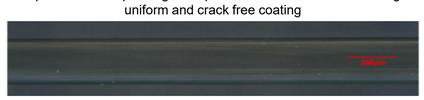
Raman spectra of thin films coated on silicon wafer before and after 1 week exposure in seawater showing stability



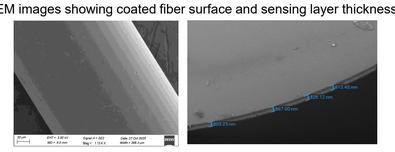
Optical fiber based pH sensing

Sol-gel synthesis → Dip coating on optical fiber → Thermal curing at 100°C for 1h

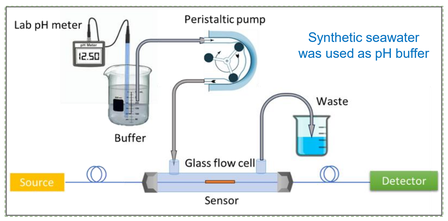
Optical microscope image of dip coated multimode fiber showing uniform and crack free coating



SEM images showing coated fiber surface and sensing layer thickness



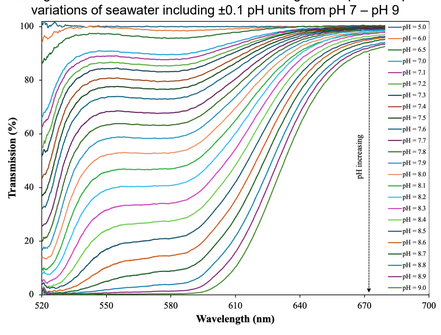
pH sensing flow cell reactor setup for evanescent wave absorption based fiber optic pH sensor



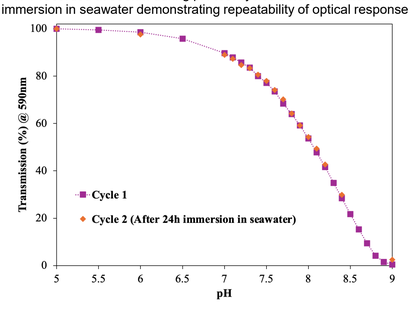
Sensing layer coated optical fiber was conditioned in seawater overnight before starting pH sensing tests.

- Preconditioning in weak alkaline solution improves ionic equilibrium and removes any loosely bound species in the layer as well as minimizes and prevents re-formation of water molecules and hydrogen bonds within the sol-gel matrix
- Proton uptake and release occurs more rapidly and reversibly after preconditioning

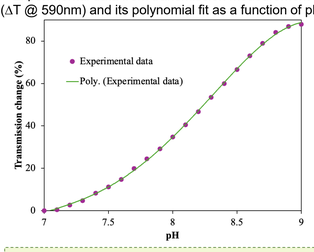
Change in transmission as a function of wavelength in response to pH variations of seawater including ±0.1 pH units from pH 7 – pH 9



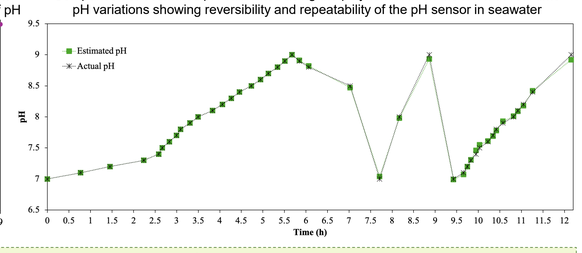
% T recorded at 590nm during pH test cycles before and after 24h immersion in seawater demonstrating repeatability of optical response



Experimental data from transmission measurement (ΔT @ 590nm) and its polynomial fit as a function of pH



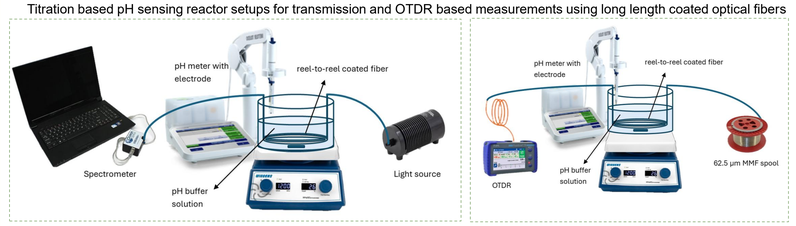
Comparison of estimated pH calibrated using the polynomial fit on the left with real time pH variations showing reversibility and repeatability of the pH sensor in seawater



mCP dye immobilized organically modified silane coated optical fiber pH sensor demonstrated < ±0.1pH sensitivity in seawater with sensor response time of ~3 minutes for forward and ~6 minutes for reverse pH unit variations

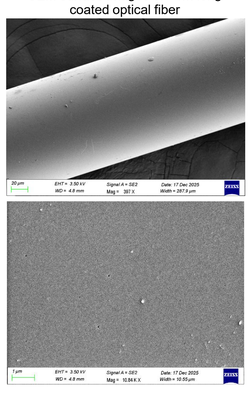
Distributed sensing feasibility demonstrations

Titration based pH sensing reactor setups for transmission and OTDR based measurements using long length coated optical fibers

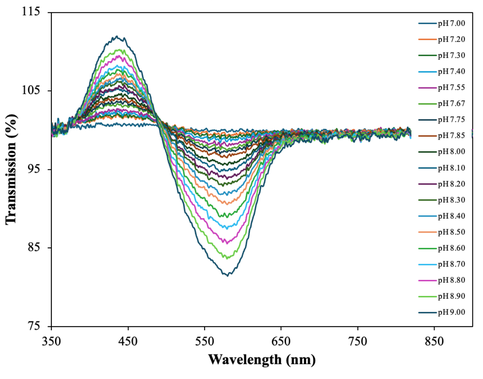


mCP dye immobilized organically modified silica layer optimized from point sensor benchmarking was synthesized and successfully coated over >1m length on optical fiber using a reel-to-reel coating system

SEM surface images of >1m long coated optical fiber



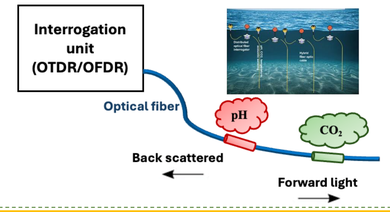
Transmission changes of 1m long coated fiber with respect to 0.1pH variations



Future directions:

- Optically sensitized and modified silica with NIR absorption for feasibility with commercial OTDR wavelengths (850nm or 1300nm)
- Sensing layer optimization for pH sensing precision of < ±0.05 pH units along with compatibility and long term stability in marine environment

Distributed fiber optic interrogation for chemical sensing in marine environment



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